Session 18 (AIML) - Assignment Question

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- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML
- **Github url:** https://github.com/shrutidarwatkar-sketch/Task_ITR.git

Q1. Data Loading & Initial Analysis) Load the Ford Car Dataset using pandas.

- Display the first 10 rows and last 5 rows.
- Check the shape (number of rows and columns) and data types of all columns.
- Write observations about the dataset structure in comments

PROGRAM SCREENSHOT :

```
]
0s
    import pandas as pd

]
9s
    from google.colab import files
    uploaded = files.upload()

    Choose files insurance.csv
    insurance.csv(text/csv) - 50264 bytes, last modified: 09/07/2026 - 100% done
    Saving insurance.csv to insurance.csv
```

```python
#Q1

# Load the dataset
df = pd.read_csv('/content/insurance.csv')

# Display first 10 rows
print("First 10 Rows")
print(df.head(10))

# Display last 5 rows
print("\nLast 5 Rows")
print(df.tail())

# Display shape
print("\nShape of Dataset:")
print(df.shape)

# Display data types
print("\nData Types:")
print(df.dtypes)

# Observations:
# 1. The dataset contains both numerical and categorical columns.
# 2. Each row represents an insurance customer.
# 3. 'charges' is the target variable.
```

```
First 10 Rows
   age     sex   bmi  children smoker     region  expenses
0   19  female  27.9         0    yes  southwest  16884.92
1   18    male  33.8         1     no  southeast   1725.55
2   28    male  33.0         3     no  southeast   4449.46
3   33    male  22.7         0     no  northwest  21984.47
4   32    male  28.9         0     no  northwest   3866.86
5   31  female  25.7         0     no  southeast   3756.62
6   46  female  33.4         1     no  southeast   8240.59
7   37  female  27.7         3     no  northwest   7281.51
8   37    male  29.8         2     no  northeast   6406.41
9   60  female  25.8         0     no  northwest  28923.14

Last 5 Rows
      age     sex   bmi  children smoker     region  expenses
1333   50    male  31.0         3     no  northwest  10600.55
1334   18  female  31.9         0     no  northeast   2205.98
1335   18  female  36.9         0     no  southeast   1629.83
1336   21  female  25.8         0     no  southwest   2007.95
1337   61  female  29.1         0    yes  northwest  29141.36

Shape of Dataset:
(1338, 7)

Data Types:
age           int64
sex          object
bmi         float64
children      int64
smoker       object
region       object
expenses    float64
dtype: object
```

Q2. Missing & Duplicate Values) Perform a detailed check on the dataset:

- Find the total number of missing values in each column.
- Identify and remove duplicate rows if any.
- Write a summary of your findings and how you handled them in comments.

PROGRAM SCREENSHOT :

```
#Q2

# Check missing values
print("Missing Values:")
print(df.isnull().sum())

# Check duplicate rows
print("\nDuplicate Rows:", df.duplicated().sum())

# Remove duplicate rows
df = df.drop_duplicates()

print("\nShape After Removing Duplicates:")
print(df.shape)

# Observation:
# Duplicate rows are removed if present.
# Missing values should be handled before model building.
```

```
Missing Values:
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
expenses    0
dtype: int64

Duplicate Rows: 1

Shape After Removing Duplicates:
(1337, 7)
```

Q3. (Statistical Summary)

Generate the statistical summary of all numeric columns using describe().

- Identify minimum, maximum, mean, and median values for key columns like price, mileage, and year.

PROGRAM SCREENSHOT :

```
#Q3
# Statistical Summary

print(df.describe())

# Minimum, Maximum, Mean and Median
columns = ['age', 'bmi', 'children', 'expenses']

for col in columns:
    print(f"\nColumn: {col}")
    print("Minimum :", df[col].min())
    print("Maximum :", df[col].max())
    print("Mean    :", df[col].mean())
    print("Median  :", df[col].median())
```

```
                age          bmi     children      expenses
count  1338.000000  1338.000000  1338.000000   1338.000000
mean     39.207025    30.665471     1.094918  13270.422414
std      14.049960     6.098382     1.205493  12110.011240
min      18.000000    16.000000     0.000000   1121.870000
25%      27.000000    26.300000     0.000000   4740.287500
50%      39.000000    30.400000     1.000000   9382.030000
75%      51.000000    34.700000     2.000000  16639.915000
max      64.000000    53.100000     5.000000  63770.430000

Column: age
Minimum : 18
Maximum : 64
Mean    : 39.20702541106129
Median  : 39.0

Column: bmi
Minimum : 16.0
Maximum : 53.1
Mean    : 30.66547085201794
Median  : 30.4

Column: children
Minimum : 0
Maximum : 5
Mean    : 1.0949177877429
Median  : 1.0

Column: expenses
Minimum : 1121.87
Maximum : 63770.43
Mean    : 13270.422414050823
Median  : 9382.029999999999
```

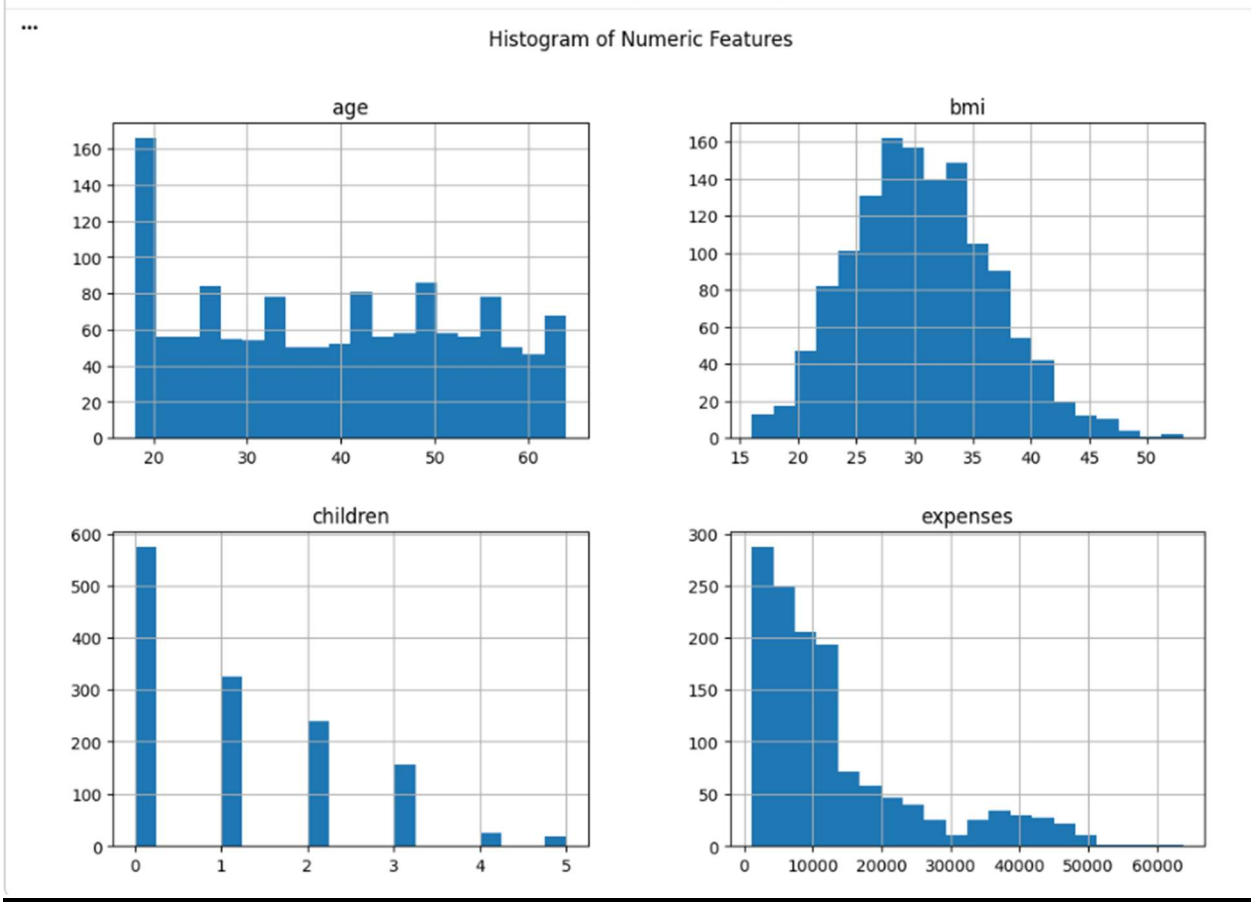
Q4. Histogram of Numeric Features)

Plot histograms for all important numeric columns (price, mileage, year, engine Size, mpg).

- Analyze the distribution for each feature and note your findings in comments.

PROGRAM SCREENSHOT :

```
#Q4

import matplotlib.pyplot as plt

numeric_columns = ['age', 'bmi', 'children', 'expenses']

df[numeric_columns].hist(figsize=(12,8), bins=20)

plt.suptitle("Histogram of Numeric Features")
plt.show()

# Observation:
# Histograms show the distribution of numerical variables.
```



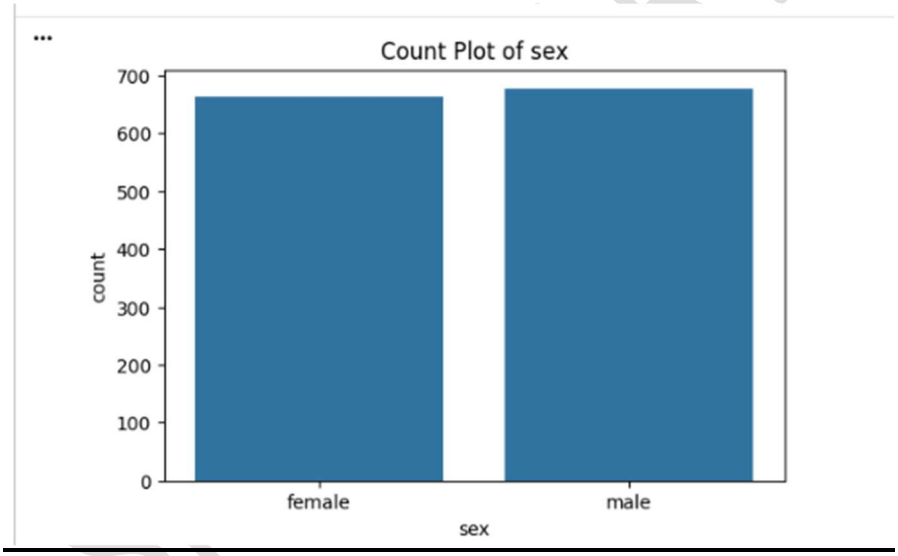
Q5. Count Plots of Categorical Features)

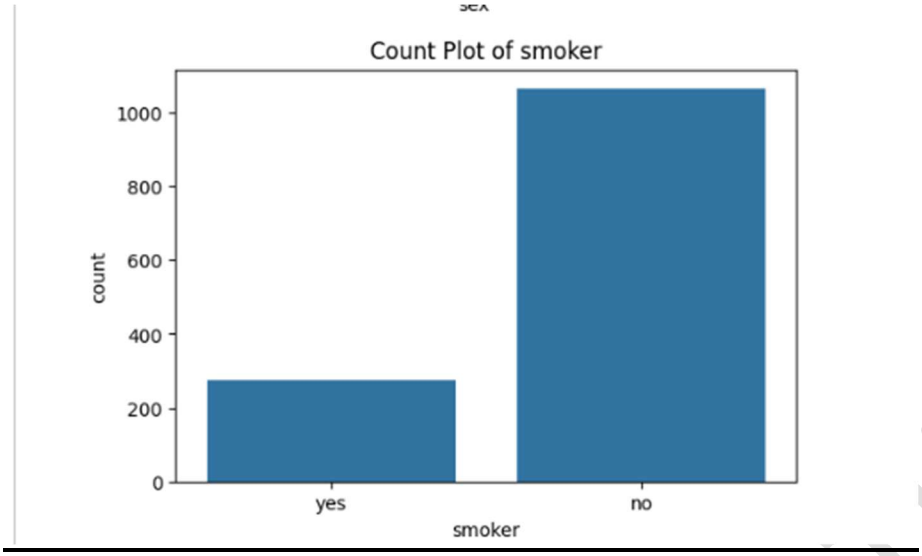
Create count plots for all categorical columns (fuel Type, transmission, model, etc.) using seaborn.

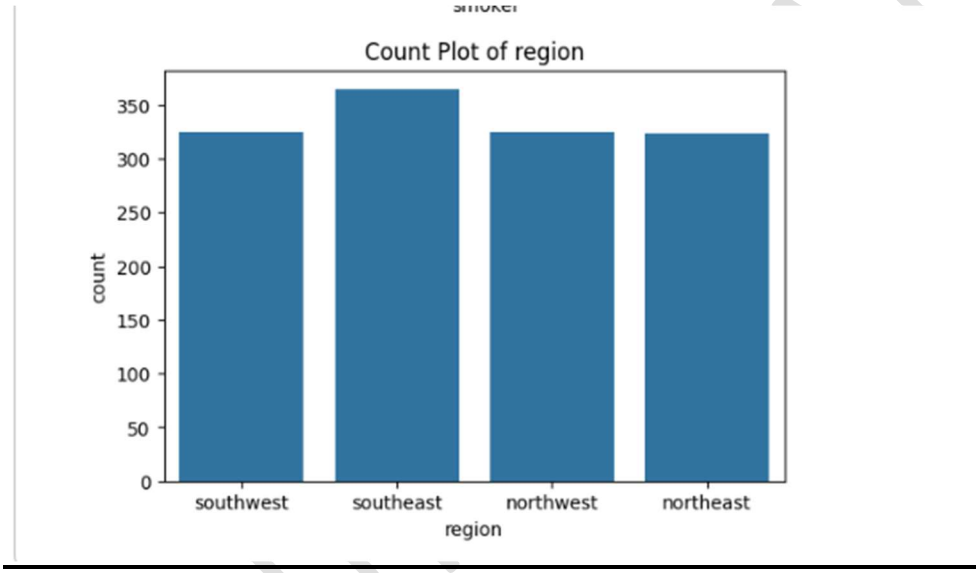
- Analyze which categories are most common.
- Write insights about market trends based on these plots in comments.

PROGRAM SCREENSHOT :

```
#Q5
import seaborn as sns
import matplotlib.pyplot as plt

categorical_columns = ['sex', 'smoker', 'region']

for col in categorical_columns:
    plt.figure(figsize=(6,4))
    sns.countplot(data=df, x=col)
    plt.title(f"Count Plot of {col}")
    plt.show()

# Observation:
# Count plots show the frequency of each category.
```




Count Plot of smoker
count
1000
800
600
400
200
0
yes
no
smoker


Count Plot of region
count
350
300
250
200
150
100
50
0
southwest
southeast
northwest
northeast
region

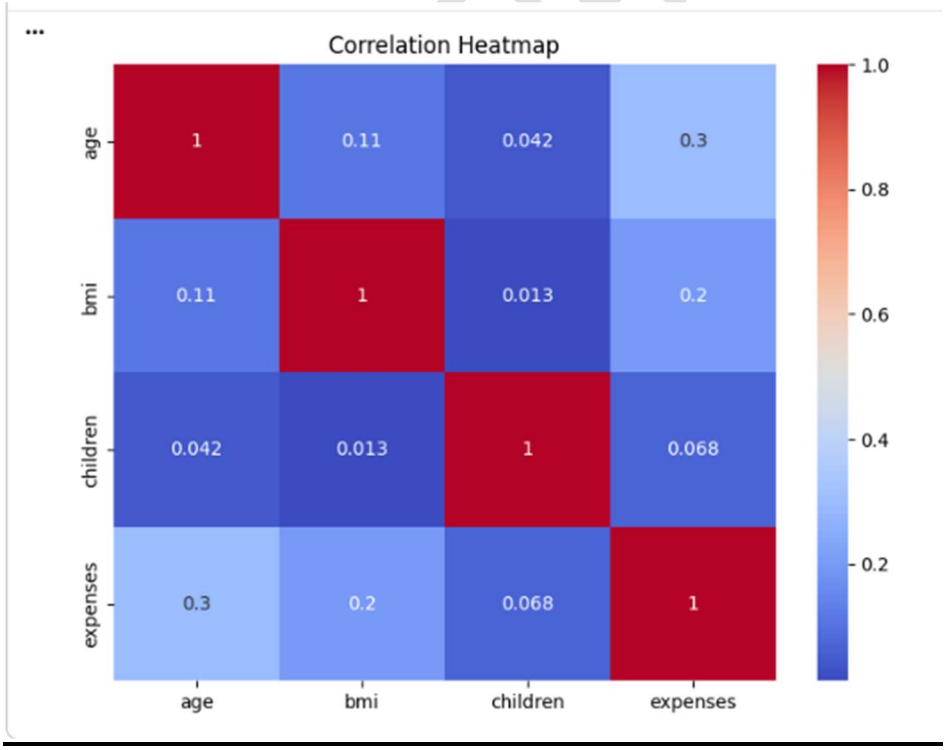
Q6. Correlation Heatmap)

Create a heatmap of the correlation matrix for numeric features.

- Use seaborn with proper annotations.
- Identify which features are highly correlated with the target variable (price).
- Write your observations in comment.

PROGRAM SCREENSHOT :

```
#Q6

import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))

sns.heatmap(df.corr(numeric_only=True),
            annot=True,
            cmap='coolwarm')

plt.title("Correlation Heatmap")
plt.show()

# Observation:
# Identify features highly correlated with expenses.
```



Q7. Feature Identification) Clearly identify and list:

- Independent Features Input Features)
- Dependent Feature Output/Target Feature)

Justify your choice with reasoning.

PROGRAM SCREENSHOT :

```
#Q7

# Independent Features
X = df.drop('expenses', axis=1)

# Dependent Feature
y = df['expenses']

print("Independent Features:")
print(X.columns)

print("\nDependent Feature:")
print(y.name)

# Justification:
# 'expenses' is the output variable.
# All remaining columns are input features.
```

```
Independent Features:
Index(['age', 'sex', 'bmi', 'children', 'smoker', 'region'], dtype='object')

Dependent Feature:
expenses
```

Q8. Encoding Categorical Variables)

Convert all categorical columns into numeric form using appropriate encoding techniques:

- Use One Hot Encoding.
- Show before and after transformation for at least 2 columns.

PROGRAM SCREENSHOT :

```
#Q8

# Display Before Encoding
print("Before Encoding")
print(df[['sex','smoker','region']].head())

# One Hot Encoding
encoded_df = pd.get_dummies(df, drop_first=True)

print("\nAfter Encoding")
print(encoded_df.head())

# Observation:
# Categorical variables are converted into numeric variables.
```

```
Before Encoding
      sex smoker     region
0  female    yes  southwest
1    male     no  southeast
2    male     no  southeast
3    male     no  northwest
4    male     no  northwest

After Encoding
   age   bmi  children  expenses  sex_male  smoker_yes  region_northwest  \
0   19  27.9         0  16884.92     False        True             False
1   18  33.8         1   1725.55      True       False             False
2   28  33.0         3   4449.46      True       False             False
3   33  22.7         0  21984.47      True       False              True
4   32  28.9         0   3866.86      True       False              True

   region_southeast  region_southwest
0             False             True
1              True            False
2              True            False
3             False            False
4             False            False
```

Q9. Feature Scaling)

Apply Standard Scaling (using StandardScaler from sklearn) on the numeric independent features

- . Show the first 5 rows of scaled data.

PROGRAM SCREENSHOT :

```
#Q9

from sklearn.preprocessing import StandardScaler

# Input Features
X = df.drop('expenses', axis=1)

# One Hot Encoding
X = pd.get_dummies(X, drop_first=True)

# Numerical Columns
numeric_columns = ['age', 'bmi', 'children']

# Standard Scaling
scaler = StandardScaler()

X[numeric_columns] = scaler.fit_transform(X[numeric_columns])

print("Scaled Data")
print(X.head())
```

```
Scaled Data
        age       bmi  children  sex_male  smoker_yes  region_northwest  \
0 -1.438764 -0.453646 -0.908614     False        True             False
1 -1.509965  0.514186 -0.078767      True       False             False
2 -0.797954  0.382954  1.580926      True       False             False
3 -0.441948 -1.306650 -0.908614      True       False              True
4 -0.513149 -0.289606 -0.908614      True       False              True

   region_southeast  region_southwest
0             False              True
1              True             False
2              True             False
3             False             False
4             False             False
```

Q10. Mini Project - Complete Preprocessing Pipeline)

Create a complete preprocessing pipeline for the Ford Car Dataset:

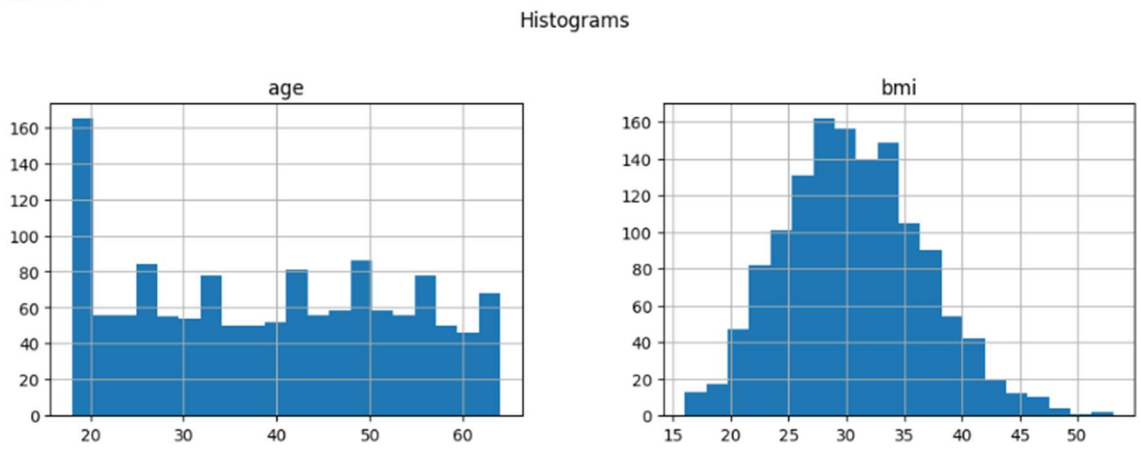
1. Load and clean the data (handle missing & duplicate values).
2. Perform EDA (histograms + count plots + heatmap).
3. Identify input and output features.
4. Encode categorical variables.

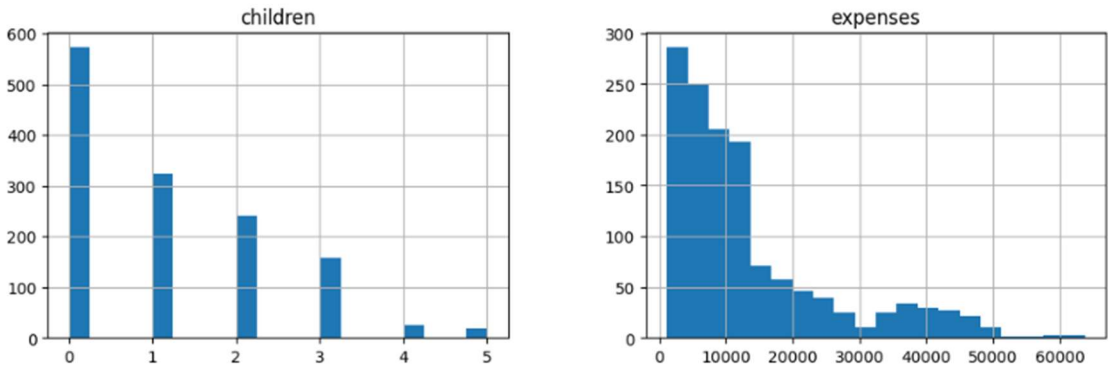
PROGRAM SCREENSHOT :

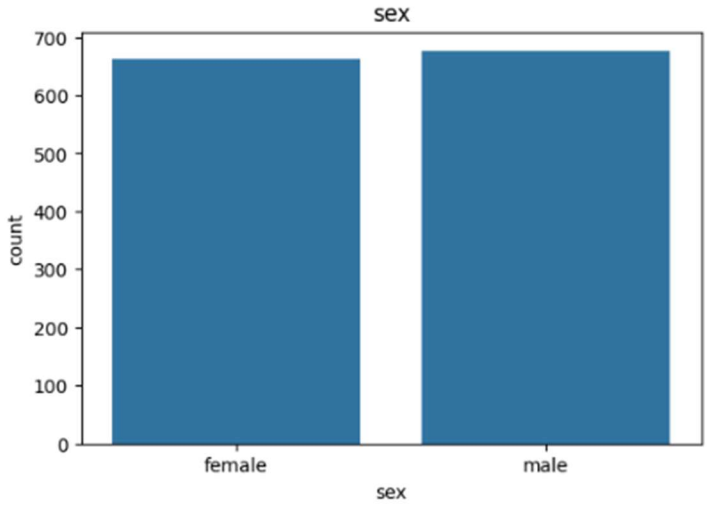
```
#Q10

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler

# Load Dataset
data = pd.read_csv('/content/insurance.csv')

# Remove Duplicates
data.drop_duplicates(inplace=True)

# Missing Values
print(data.isnull().sum())

# Histograms
numeric_columns = ['age', 'bmi', 'children', 'expenses']

data[numeric_columns].hist(figsize=(12,8), bins=20)
plt.suptitle("Histograms")
plt.show()

# Count Plots
categorical_columns = ['sex', 'smoker', 'region']

for col in categorical_columns:
    plt.figure(figsize=(6,4))
    sns.countplot(data=data, x=col)
    plt.title(col)
    plt.show()

# Correlation Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(data.corr(numeric_only=True),
            annot=True,
            cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

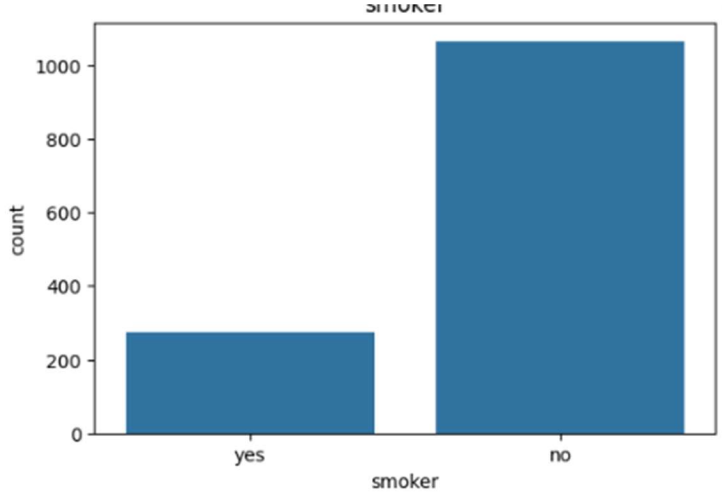
```
plt.show()

# Input and Output Features
X = data.drop('expenses', axis=1)
y = data['expenses']

# One Hot Encoding
X = pd.get_dummies(X, drop_first=True)

# Feature Scaling
numeric_columns = ['age', 'bmi', 'children']

scaler = StandardScaler()

X[numeric_columns] = scaler.fit_transform(X[numeric_columns])

print("Processed Features")
print(X.head())

print("\nTarget Variable")
print(y.head())

print("\nPreprocessing Completed Successfully.")
```

```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
expenses    0
dtype: int64
```

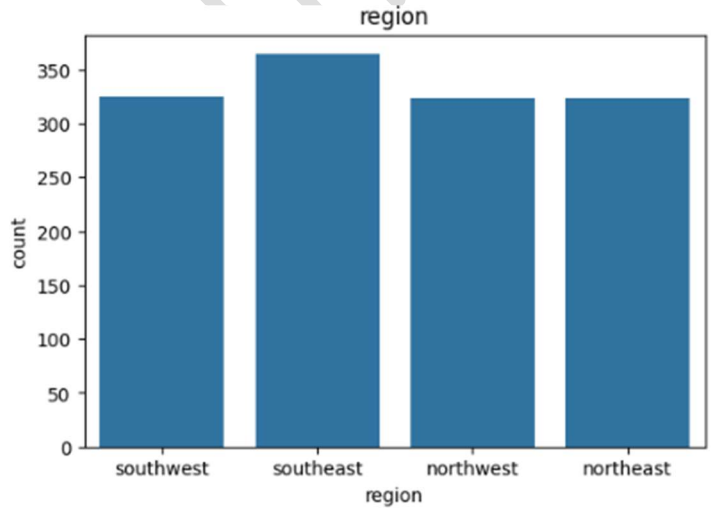


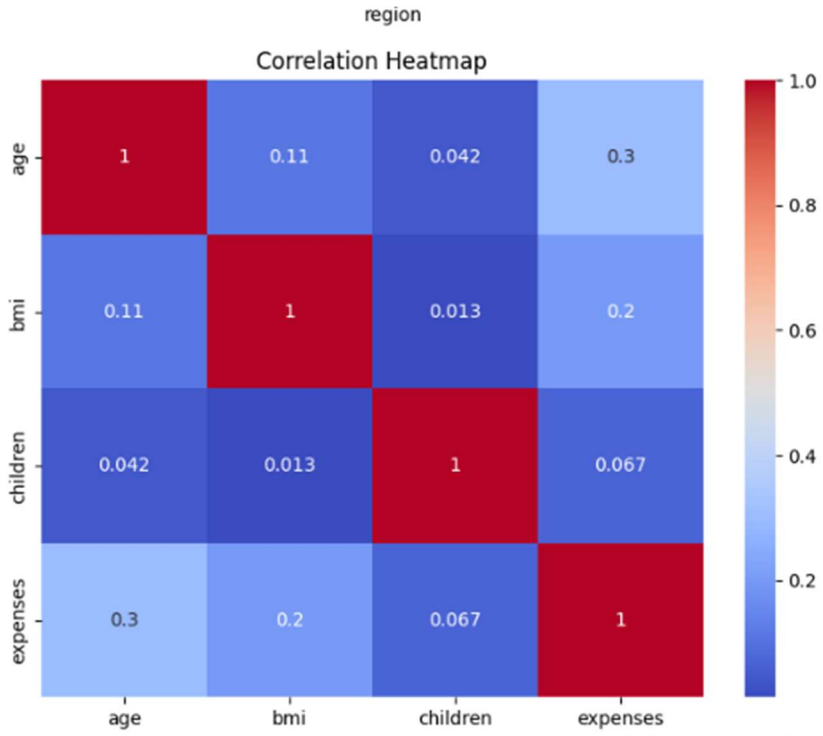



sex
700
600
500
400
300
200
100
0
count
female
male
sex


1000
800
600
400
200
0
count
yes
no
smoker

region


region
350
300
250
200
150
100
50
0
count
southwest
southeast
northwest
northeast
region




```
          age       bmi
Processed Features
        age       bmi  children  sex_male  smoker_yes  region_northwest  \
0 -1.440418 -0.453484 -0.909234     False        True             False
1 -1.511647  0.513986 -0.079442      True       False             False
2 -0.799350  0.382803  1.580143      True       False             False
3 -0.443201 -1.306169 -0.909234      True       False              True
4 -0.514431 -0.289506 -0.909234      True       False              True

   region_southeast  region_southwest
0             False              True
1              True             False
2              True             False
3             False             False
4             False             False

Target Variable
0    16884.92
1     1725.55
2     4449.46
3    21984.47
4     3866.86
Name: expenses, dtype: float64

Preprocessing Completed Successfully.
```